

U.S. EV registrations vs. energy cleanliness (2023)

Overview

“The main research question of this project is: does the electricity used to charge electric vehicles (EVs) really come from clean energy? To explore this, this report will focus on a more specific question: are EV registrations concentrated in states with cleaner energy mixes? This analysis will use 2023 data for comparison.”

Part 0: libraries

Part 1: Defining Research Question

“Are EV registrations concentrated in states with cleaner energy mixes?”

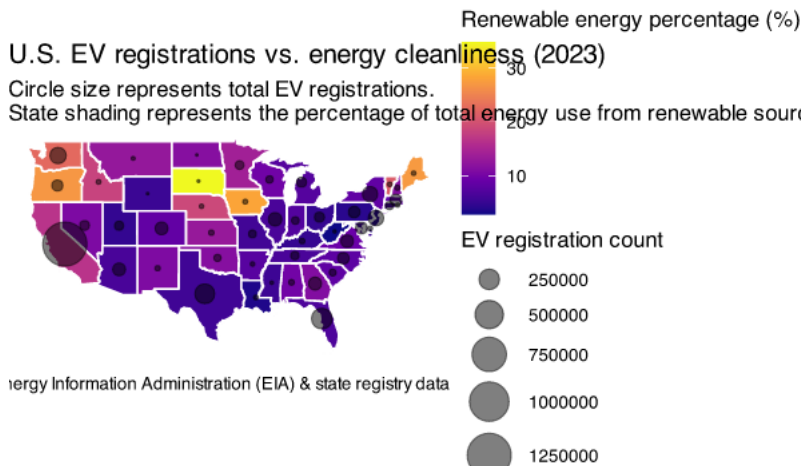
Part 2: Data Preparation and Cleaning

Part 3: Joining / Pivoting Datasets for Analysis

```
# A tibble: 6 × 7
  state_abb year total_usage renewable_usage percent_renewable state
count_evs
  <chr>      <chr>      <dbl>          <dbl>          <dbl> <chr>
<dbl>
1 AK        2023        746979        10087          1.35 Alaska
2697
2 AL        2023       2265008       222189          9.81 Alaba...
13047
3 AR        2023       1151062         87277          7.58 Arkan...
7108
4 AZ        2023       1712667       108445          6.33 Arizo...
89798
5 CA        2023       6429818      1065179         16.6 Calif...
1256646
6 CO        2023       1359507       115061          8.46 Color...
```

Part 4: Mapping Visualization

“The figure below shows our analysis results. The base color of each state represents its percentage of renewable energy (the brighter the color, the higher the percentage), while the size of the circle at the center of each state represents the state’s total EV registrations.”



“California (CA): It’s a huge circle, and the background color is very bright. This supports the view that”EVs are popular in clean-energy states.”

Texas (TX) or Florida (FL): They also have large circles, but their background colors is very dark (low share of renewable energy). This contradicts that view.”

“The analysis is complex. On one hand, California — which has the highest number of EV registrations — has a relatively high share of renewable energy. On the other hand, other states with large EV registrations (such as Texas and Florida) have relatively low shares of renewables. This suggests that although EVs are popular in some clean-energy states, many EVs nationwide still rely on grids powered by non-clean energy.”